

Interview Experience – RAMCO Systems

Key Preparation Tips (Summary)

- **Prepare all standard problems from LeetCode**, especially data structures such as linked lists, binary trees, stacks, and two-pointer/string problems.
 - **Prefer Striver's SDE Sheet** for structured DSA preparation.
 - **Understand all core Computer Science concepts** including OS, DBMS, CN, OOPs, and SQL thoroughly.
 - **Be confident and communicate clearly** – strong articulation helps significantly during both technical rounds.
 - With proper preparation and confidence, **the interview process is very crackable**.
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Introduction

This document summarizes my **complete interview experience at RAMCO Systems**, including all rounds, questions asked, and my overall performance. The entire selection process consisted of multiple filtering stages, including assessments, technical written tests, and two in-depth technical interviews conducted at CUIC.

1. Online Assessment Round

The process began with an **online assessment** that focused mainly on coding and basic aptitude.

I was required to solve **two coding problems**, both heavily centered on **Linked Lists** (operations, traversal, and edge-case handling). The round also included a set of **MCQs** related to programming fundamentals, computer science concepts, and logical reasoning. I performed well and progressed to the next stage.

2. Offline Written Technical Test

A total of **73 candidates** were shortlisted for the Written Technical Test.

We were given **three coding problems**, and ample time to write complete solutions, including optimized logic and edge-case handling.

Questions Asked

1. **Longest Non-Repeating Substring**
2. **Finding All Duplicate Subtrees** in a Binary Tree
3. **Longest Harmonic Subsequence**

This round tested our ability to write structured, clean, and optimized code. My solutions were well-received, and I was shortlisted for the first technical interview.

3. Offline Technical Interview – Round 1 (CUIC)

A total of **42 candidates** advanced to Technical Round 1.

This round involved detailed analysis of the answers written in the previous coding test as well as conceptual depth in DSA and SQL.

Questions & Topics Covered

- **Optimizing my written test solutions**
- Checking **whether two binary trees are identical**
- **Implementing a Queue using two Stacks**
- **Detecting a loop in a linked list**
- Writing an **SQL query using LIMIT** and explaining execution order
- A detailed discussion about **all my projects**:
 - Tech stacks used
 - Architecture

- Challenges faced
- Problem-solving approach
- A short **System Design task** for a *Vehicle Enterprise Management System*
- Conversation about **extracurricular activities and leadership roles**

This round went very well, my explanations were strong, and I answered all the questions confidently. I expected to progress to the next round—and I did.

4. Offline Technical Interview – Round 2 (CUIC)

Only **20 candidates** were shortlisted for this deeper technical evaluation.

This round primarily focused on **project depth, CS fundamentals, and problem-solving skills**.

Topics Covered

Project Deep-Dive

They asked about:

- Complete project architecture
- Every tech stack I used
- “Why” questions such as: **Why MongoDB over PostgreSQL?**
- Data structures used in each module
- Future scope and improvements
- Real-world scalability considerations

Coding and Problem-Solving

1. A **recursion program** where I had to explain the output and derive **time complexity**
2. A **Binary String problem** requiring pattern analysis

Core CS Concepts

A thorough discussion of:

- **OOPs Concepts** (Inheritance, Polymorphism, Abstraction, Encapsulation)
- **Operating Systems** (deadlocks, scheduling, memory)
- **DBMS** (ACID, indexing, joins)
- **Computer Networks** (OSI layers, protocols)

This round was challenging but went smoothly. I answered confidently and showcased a strong understanding of both my projects and CS fundamentals.

Final Shortlist

After all rounds, **10 candidates were finally selected** for the final stage.

The interviewers emphasized confidence, problem-solving approach, and conceptual understanding as key differentiators.

Best Wishes,
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